

JEE MAIN REVISION MASTER: CHEMICAL BONDING & MOLECULAR STRUCTURE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 214 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Chemical Bonding And Molecular Structure - Core Application	214 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Chemical Bonding And Molecular Structure - Core Application (Very High Yield — 214 PYQs)

Core Principle: Key problem pattern in Chemical Bonding And Molecular Structure.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Bromine trifluoride autoionizes to form BrF_2^+ and BrF_4^- . The shapes of the cation and anion are respectively ____ and..." [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "The covalent radii of atoms A and B are r_A and r_B , respectively. The covalent bond length and total length of AB molecule are..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Match List - I with List - II according to shape. List - I List - II A. XeO_3 B. XeF_2 C. XeO_2 D. XeOF_4 E. BrF_5 F. NH_3 ... " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Bromine trifluoride autoionizes to form BrF_2^+ and BrF_4^- . The shapes of the cation and anion are respectively _____ and _____.

- (A) bent, square planar **[Correct]**
- (B) linear, square planar
- (C) bent, see-saw
- (D) linear, tetrahedral

Correct Answer: A

Solution Analysis: First, let us recall the autoionization of bromine trifluoride: $2\text{BrF}_3 \rightleftharpoons \text{BrF}_2^+ + \text{BrF}_4^-$. We have to determine the shapes of the cation and the anion. Central atom: Br Valence electrons of Br = 7 Each F contributes one bond $\rightarrow 2 \text{ Br-F bonds} = 2 \text{ pairs of bonding electrons}$ So, total electrons around Br for shape determination: $7 - 2(\text{by bonding}) - 1(\text{positive charge}) = 4$ These 4 electrons form 2 lone pairs on Br. Hence, the total number of electron pairs around Br = $2 (\text{bonding}) + 2 (\text{lone pairs}) = 4$ electron domains. According to VSEPR theory: 4 electron pairs $\rightarrow$